

07/28/26

Hardware Requirements

Tier	REQ ID	Parent ID	Requirement Text	Verification
1	MFD-HW-001	-	The MFD Display Unit shall provide the hardware computing, supervisory control, aircraft-data interface, memory, storage, network, removable-media, display, power-management, reset, and watchdog	Inspection
2	MFD-HW-100	MFD-HW-001	The MFD Display Unit shall provide a Linux application and graphics processing function.	Test
3	MFD-HW-101	MFD-HW-100	The Linux application and graphics processing function shall use one AMD XQZU3EG-1SFRA484I XQ Zynq UltraScale+ MPSoC.	Inspection
3	MFD-HW-102	MFD-HW-100	The selected MPSoC hardware shall provide application-processing, GPU, and FPGA-fabric resources.	Inspection + Demonstration
3	MFD-HW-103	MFD-HW-100	The MPSoC implementation shall provide interfaces for DDR4, QSPI, eMMC, DisplayPort-class display output, and Ethernet	Inspection + Test
3	MFD-HW-104	MFD-HW-100	The MPSoC shall be installed in the 484-ball FCBGA package defined by the selected component.	Inspection
2	MFD-HW-110	MFD-HW-001	The MFD Display Unit shall provide an independent RTOS/supervisory processing function.	Test
3	MFD-HW-111	MFD-HW-110	The supervisory processing function shall use one Texas Instruments S5LS20216ASPGEMEP TMS570LS20216-EP Hercules MCU with lockstep Cortex-R4F processing.	Inspection
3	MFD-HW-112	MFD-HW-110	The supervisory MCU implementation shall provide SPI, UART, JTAG, and GPIO connectivity required by the MFD hardware architecture.	Inspection + Test
3	MFD-HW-113	MFD-HW-110	The supervisory MCU shall be electrically independent of the Linux application processor such that supervisory functions are not dependent on Linux execution.	Inspection + Test
2	MFD-HW-120	MFD-HW-001	The MFD Display Unit shall provide ARINC 429 aircraft-data communications.	Test
3	MFD-HW-121	MFD-HW-120	The ARINC 429 interface shall use two Holt HI-35930PQTF devices.	Inspection
3	MFD-HW-122	MFD-HW-120	The two HI-35930PQTF devices shall provide an aggregate capability of four ARINC 429 receive channels and two ARINC 429 transmit channels.	Inspection + Test
3	MFD-HW-123	MFD-HW-120	The ARINC 429 devices shall communicate with the supervisory MCU through SPI.	Inspection + Test

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3	MFD-HW-124	MFD-HW-120	The aircraft external I/O connector implementation shall route the required ARINC 429 differential signals between the MFD and aircraft wiring.	Inspection + Continuity Test
2	MFD-HW-130	MFD-HW-001	The MFD Display Unit shall provide DDR4 system memory for the MPSoC.	Test
3	MFD-HW-131	MFD-HW-130	The DDR4 memory implementation shall use two ISSI IS46QR16512A-075VBLA2 devices, each rated as 8-Gbit x16 DDR4 SDRAM.	Inspection
3	MFD-HW-132	MFD-HW-130	The two DDR4 devices shall provide a combined 16-Gbit memory capacity and a combined x32 data-width implementation.	Inspection + Test
2	MFD-HW-140	MFD-HW-001	The MFD Display Unit shall provide nonvolatile boot/configuration storage.	Test
3	MFD-HW-141	MFD-HW-140	Boot/configuration storage shall use one Micron MT25QL512ABB8E12-0AUT 512-Mbit Quad-SPI NOR flash device.	Inspection
3	MFD-HW-142	MFD-HW-140	The QSPI boot-flash interface shall operate from the 3.3 V interface requirement identified for the selected flash component.	Inspection + Test
2	MFD-HW-150	MFD-HW-001	The MFD Display Unit shall provide main nonvolatile storage for the Linux processing environment.	Test
3	MFD-HW-151	MFD-HW-150	Main nonvolatile storage shall use one Swissbit SFEM020GB2ED1TB-A-CE-11P-STD 20 GB eMMC device.	Inspection
3	MFD-HW-152	MFD-HW-150	The eMMC device shall interface with the MPSoC through the eMMC interface.	Inspection + Test
2	MFD-HW-160	MFD-HW-001	The MFD Display Unit shall provide controlled multi-rail power for the processor and FPGA power rails.	Test
3	MFD-HW-161	MFD-HW-160	The processor power-management function shall use one Texas Instruments TPS65086401RSKT multi-rail PMIC.	Inspection
3	MFD-HW-162	MFD-HW-160	The PMIC shall provide I2C-accessible power-management control to the MFD electronics.	Inspection + Test
3	MFD-HW-163	MFD-HW-160	The MFD design shall include the external passive and support components required to implement the PMIC power rails defined by the final schematic.	Inspection
2	MFD-HW-170	MFD-HW-001	The MFD Display Unit shall provide hardware reset supervision.	Test
3	MFD-HW-171	MFD-HW-170	Hardware reset supervision shall use one Texas Instruments TPS3808G01MDBVTEP adjustable voltage supervisor/reset controller.	Inspection
3	MFD-HW-172	MFD-HW-170	The reset supervisor shall provide a hardware reset output to the supervised MFD processing circuitry when the configured monitored condition is not satisfied.	Test
2	MFD-HW-180	MFD-HW-001	The MFD Display Unit shall provide an independent hardware watchdog function.	Test

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3	MFD-HW-181	MFD-HW-180	The independent watchdog function shall use one Texas Instruments TPS3431SDRBR window watchdog timer.	Inspection
3	MFD-HW-182	MFD-HW-180	The watchdog shall detect processor servicing outside the configured watchdog window and provide the associated hardware fault response defined by the final schematic.	Test
2	MFD-HW-190	MFD-HW-001	The MFD Display Unit shall provide a Gigabit Ethernet physical interface.	Test
3	MFD-HW-191	MFD-HW-190	The Ethernet physical interface shall use one Texas Instruments DP83867IRPAPR 10/100/1000 Ethernet PHY.	Inspection
3	MFD-HW-192	MFD-HW-190	The Ethernet PHY shall interface to the MPSoC through RGMII.	Inspection + Test
3	MFD-HW-193	MFD-HW-190	The Ethernet external interface shall route the required Ethernet differential pairs through the final connector and magnetics implementation.	Inspection + Test
2	MFD-HW-200	MFD-HW-001	The MFD Display Unit shall provide a removable microSD memory-card interface.	Demonstration
3	MFD-HW-201	MFD-HW-200	The removable-media interface shall use one Molex 104031-0811 surface-mount microSD connector.	Inspection
3	MFD-HW-202	MFD-HW-200	The microSD connector shall interface to the MFD electronics using SD/MMC signaling.	Inspection + Test
3	MFD-HW-203	MFD-HW-200	The microSD signal lines shall be protected by one Texas Instruments TPD6E05U06RVZR six-channel low-capacitance ESD protection array.	Inspection
3	MFD-HW-204	MFD-HW-200	The microSD ESD-protection implementation shall cover up to six protected SD signal lines as provided by the selected protection array.	Inspection + Test
2	MFD-HW-210	MFD-HW-001	The MFD Display Unit shall provide a dedicated multifunction display panel.	Demonstration
3	MFD-HW-211	MFD-HW-210	The MFD display panel shall use the SAEF SFTO1330SM-7208A3CT2 13.3-inch LCD selected in the finalized BOM.	Inspection
3	MFD-HW-212	MFD-HW-210	The MFD display panel shall provide a native resolution of 1920 x 1080 pixels.	Inspection + Test
3	MFD-HW-213	MFD-HW-210	The MFD display panel shall provide the high-brightness display characteristic specified for the selected panel.	Inspection + Test
3	MFD-HW-214	MFD-HW-210	The MFD display panel shall interface to the MFD electronics through eDP.	Inspection + Test
2	MFD-HW-220	MFD-HW-001	The MFD Display Unit shall provide the reference oscillator and clock network required by the selected MPSoC.	Inspection + Test

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3	MFD-HW-221	MFD-HW-220	Final oscillator frequencies and component values shall be selected to satisfy the MPSoC clock-input requirements used by the final schematic.	Analysis + Inspection
2	MFD-HW-230	MFD-HW-001	The MFD Display Unit shall provide a JTAG programming and hardware-debug interface.	Inspection + Demonstration
3	MFD-HW-231	MFD-HW-230	The JTAG interface shall include the connector and support circuitry required to access the applicable MFD processing devices during development and maintenance.	Inspection
2	MFD-HW-240	MFD-HW-001	The MFD Display Unit shall provide the display connector and backlight-support circuitry required by the SFTO1330SM-7208A3CT2 LCD.	Inspection + Test
2	MFD-HW-250	MFD-HW-001	The MFD Display Unit shall provide a rugged aircraft external I/O connector for aircraft data and electrical interfaces.	Inspection
3	MFD-HW-251	MFD-HW-250	The final aircraft external connector shall provide sufficient contacts for the finalized MFD aircraft I/O and power wiring defined by the schematic.	Inspection + Continuity Test
2	MFD-HW-260	MFD-HW-001	The MFD Display Unit shall include the decoupling, pull-up, pull-down, bias, and termination networks required by the selected components and interfaces.	Inspection
3	MFD-HW-261	MFD-HW-260	Final passive values and quantities shall be established from the selected-component datasheets, reference designs, and completed schematic.	Analysis + Inspection